

Companion XXXVI: The Baryonic Cycle in the Relaxation-Driven Cyclic Cosmology

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Abstract

The formation of galaxies is governed by the complex interplay between gas cooling, star formation, and feedback. These processes form a self-regulating baryonic cycle that determines the luminosity, morphology, and chemical evolution of galaxies. In the Relaxation-Driven Cyclic Cosmology (RDCC), the scale-dependent evolution of the gravitational potential modifies the thermodynamic pathways through which gas accretes, cools, and condenses into stars. This Companion develops a continuous description of the baryonic cycle in RDCC, deriving how the relaxation scale influences cooling times, feedback efficiency, and the equilibrium between inflows and outflows. The resulting framework provides a natural explanation for the suppressed star formation in low-mass halos and the early assembly of massive galaxies observed in deep surveys.

1 Introduction

The baryonic cycle is the engine of galaxy formation. Gas flows into halos along filaments, cools radiatively, condenses into molecular clouds, forms stars, and is then reheated or expelled by stellar and AGN feedback. This cycle is not linear but self-regulating: the rate at which gas cools determines the star formation rate, which in turn determines the strength of feedback, which then modifies the cooling rate.

In RDCC, the gravitational potential that shapes this cycle evolves differently from the standard cosmological model. The relaxation mechanism suppresses the growth of small-scale modes and enhances the coherence of large-scale modes. This modifies the density, temperature, and entropy structure of the gas as it accretes onto halos. The baryonic cycle becomes sensitive to the relaxation scale k_{rel} , which imprints itself on the thermodynamic history of galaxies.

2 Gas Accretion and the RDCC Potential

Gas accretion onto halos is driven by the gravitational potential. In RDCC, this potential is modified by the relaxation mechanism:

$$\Phi_{\text{RDCC}}(k, a) = \Phi(k, a) \left[1 - \chi_{\Phi} \left(\frac{k}{k_{\text{rel}}} \right)^{\alpha} \right]. \quad (1)$$

The suppression of small-scale modes reduces the depth of the potential wells of low-mass halos. As a result, the infalling gas experiences a weaker compression, leading to lower virial temperatures. The virial temperature in RDCC becomes

$$T_{\text{vir, RDCC}} = T_{\text{vir}} \left[1 - \chi_T \left(\frac{k(M)}{k_{\text{rel}}} \right)^{\alpha} \right]. \quad (2)$$

This reduction in virial temperature has profound consequences for the cooling efficiency of gas in low-mass halos.

3 Cooling Pathways and Thermodynamic Evolution

Radiative cooling depends on the density and temperature of the gas. In RDCC, the modified virial temperature alters the cooling time:

$$t_{\text{cool,RDCC}} = \frac{3k_B T_{\text{vir,RDCC}}}{n_e \Lambda(T_{\text{vir,RDCC}})}, \quad (3)$$

where Λ is the cooling function.

Because $T_{\text{vir,RDCC}}$ is reduced for low-mass halos, the cooling time increases, delaying the onset of star formation. Massive halos, whose collapse is accelerated by the coherence of large-scale modes, maintain higher virial temperatures and shorter cooling times. This asymmetry naturally produces a suppressed star formation rate in dwarf galaxies and an enhanced early star formation rate in massive galaxies.

The thermodynamic evolution of the gas thus becomes a direct tracer of the relaxation scale.

4 Feedback and the Regulation of Star Formation

Feedback from supernovae and active galactic nuclei injects energy into the interstellar medium, reheating or expelling gas. The efficiency of feedback depends on the depth of the gravitational potential. In RDCC, the reduced potential depth of low-mass halos enhances the effectiveness of feedback, while the deeper potentials of massive halos remain largely unaffected.

The energy required to unbind gas from a halo of mass M is

$$E_{\text{bind,RDCC}} = \frac{1}{2} M_{\text{gas}} v_{\text{esc,RDCC}}^2, \quad (4)$$

with

$$v_{\text{esc,RDCC}} = v_{\text{esc}} \left[1 - \chi_v \left(\frac{k(M)}{k_{\text{rel}}} \right)^\alpha \right]. \quad (5)$$

This modification amplifies the impact of feedback in dwarf-scale halos, suppressing their star formation even further.

5 Equilibrium Between Inflows and Outflows

Galaxies evolve toward a quasi-equilibrium state in which inflows, outflows, and star formation balance each other. In RDCC, this equilibrium is shifted by the relaxation-modified potential. The equilibrium condition can be written as

$$\dot{M}_{\text{in}}(M) = \dot{M}_\star(M) + \dot{M}_{\text{out}}(M), \quad (6)$$

where each term carries an RDCC correction.

Low-mass halos reach equilibrium at lower stellar masses because their inflow rates are reduced and their outflow rates are enhanced. Massive halos reach equilibrium earlier and at higher stellar masses, reflecting their accelerated collapse and more efficient cooling.

This framework provides a natural explanation for the observed bimodality in galaxy populations.

6 Conclusion

The baryonic cycle in the Relaxation-Driven Cyclic Cosmology is shaped by the scale-dependent evolution of the gravitational potential. The relaxation mechanism modifies gas accretion, cooling, feedback, and the equilibrium between inflows and outflows. These effects suppress star formation

in low-mass halos, enhance early star formation in massive halos, and reshape the thermodynamic pathways through which galaxies assemble their stellar mass. The present Companion establishes the foundation for future studies of galaxy evolution, chemical enrichment, and the interaction between baryons and dark matter in RDCC.

References

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